

Lecture 20: Steady-State Behavior, Absorption

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—title: “Lecture 20: Steady-State Behavior, Absorption”author: “Jamie Haddock”format: revealjs: output-file: Lecture20_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
P = np.array([[0.8, 0.2], [0.4, 0.6]])A_eq = np.vstack([P.T - np.eye(2), np.ones(2)])b_eq = np.array([0
```

—::: {callout-note icon=false}## **Definition: Absorption**For chains with absorbing states, the probability of eventual absorption into a given state, starting from state i , satisfies a linear system obtained by conditioning on the first step.:::## **Reading & Homework***Reading: §7.3-7.4 (Bertsekas & Tsitsiklis, Introduction to Probability).***Due today:** Homework 9 (Gradescope).